

Urbanization and CO₂ Emissions in Developing Countries

MATH318 – Statistical Learning | Spring 2026

Project Report

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Name	ID	Contribution <i>Since it is a group project, we all contributed to all parts of the project. However, the main research and writing of these parts were split as follows:</i>
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1. Abstract

This study examines the relationship between urbanization and CO₂ emissions in developing countries using a panel set spanning the years 1975 through 2002. As a baseline, we implemented linear and quadratic pooled OLS models to study cross-country dynamics. This was followed by country and time fixed effect models to investigate within-country effects. Both pooled models suggested the presence of a statistically significant relationship between urbanization and CO₂ emissions, with the quadratic model providing evidence of a non-linear relationship exhibiting an inverted U-shape pattern. However, these results do not persist once country and time fixed effects are introduced. In the preferred two-way fixed effects model, results show that urbanization is not statistically significant, whereas energy consumption per capita remains a strong, robust, and statistically significant predictor of CO₂ emissions. Additional robustness checks—including first differences, log transformation, and lagged model specification—reaffirm these findings. Overall, the results indicate that the relationship observed in pooled models is driven by cross-country differences rather than within-country differences over time. This suggests that policies regarding CO₂ emissions should focus more directly on energy consumption and efficiency than on urbanization alone.

2. Introduction

In recent decades, many developing countries have experienced surges of rapid urbanization, causing impacts on factors including but not limited to economic activity, energy consumption, environmental outcomes, and population fluctuation. Although urbanization can improve a country's productivity and living conditions, it may also lead to increased CO₂ emissions due to higher energy demands. Hence, understanding this relationship is crucial for environmental and development policy-making.

This paper investigates how urbanization affects CO₂ emissions in developing countries, and discerns whether this relationship presents a nonlinear pattern and an environmental turning point. Increased focus is directed towards distinguishing cross-country and within-country urbanization and CO₂ relationships. Using a panel dataset on several countries over previous decades, we estimate pooled OLS models as well as country and time fixed effects models. Pooled models suggest a significant relationship between urbanization and emissions, whereas the preferred two-way fixed-effects model provides no evidence of such a relationship. In all models, energy consumption is consistently a strong predictor of emissions.

The remainder of the paper proceeds as follows: we begin with the data description and methodology, followed by baseline and fixed effects models, then robustness checks, and conclude with a discussion of heterogeneity and policy implications. A key contribution of this study is distinguishing between cross-country correlations and within-country dynamics using panel data methods.

3. Data & Exploratory Data Analysis (EDA)

3.1 Data Description

As mentioned above, the dataset that is being used is a panel data set which consists of observations on multiple countries across multiple years. Panel data is essentially a merge between cross-sectional data and a time series. In our case, we have balanced panel data, meaning all countries have observations across all years. There are 36 countries with 28 years of data. The data includes variables about CO₂ emissions per capita, the population in millions, energy consumption per capita, GDP per capita, all their growth rates, the lagged value of CO₂ emissions, urbanization index, carbon index, education index, and a categorical variable identifying each country. All the variables are varying regressors, meaning they vary with both the country and the year, except for the categorical variable for identifying countries, which is a time-invariant regressor, meaning it is constant across the years.

3.2 Exploratory Data Analysis

The initial analysis concludes that none of the entries in the dataset is missing, all variables are float numbers except for country, and none of the countries has any typos. There are 28 observations per country, each corresponding to a year from 1975 to 2002. Given this information, we add a year variable to the dataset.

In *Figure 1*, a scatter plot of CO₂ against urbanization for all countries showed that a trend does exist and suggests a positive relationship in the pooled data, although the spread of the observations indicates some hidden type of relationship, possibly depending on the country.

In *Figure 2*, the heatmap correlation matrix between all variables from the data set shows that CO₂ emissions are positively correlated to energy consumption and GDP, while the relationship with urbanization appears weak. This could imply that urbanization is affected by other factors that are specific to each country. Performing data analysis on the entire data set, like the heatmap, fails to account for each country's other factors/effects and masks the underlying trends. This leads to the next step, plotting a histogram for the correlation between CO₂ emissions and urbanization for each country. In *Figure 3*, we can immediately see what was hidden. In most countries, the CO₂ emissions show strong correlation, and in many of them, it is a negative relationship where CO₂ emissions decrease as urbanization increases. This confirms that the effect of urbanization is not consistent across countries.

In *Figure 4*, splitting the scatter plot of CO₂ emissions against urbanization down to the countries allowed us to see specific trends; we were able to see patterns that were hidden in the pooled scatter plot. Some of the plots show evidence for an almost linear relationship in many countries; some are positive, some are negative, and some show quadratic relationships.

As for *Figure 5 and 6*, these time-series plots show how CO₂ emissions and urbanization changes over time for each country. The plots show that they change differently across countries; some increase or decrease constantly, while several others show non-linear patterns. The consistency of such evidence of these relationships in each case shows that these variables are important features in the dataset.

Some countries had CO₂ emissions decrease as urbanization increased over time, and some had the opposite. However, only South Korea and Turkey had both increased over time. It is also good to note that for some countries, their urbanization started off high, then (between 1990 and 1995) had their CO₂ emissions decrease, with urbanization still increasing. A possible explanation for this could be technological advancements in the world that improved living conditions, meaning urbanization, and also improved the management of CO₂ emissions, meaning a decrease in them.

These findings highlight the importance of accounting for country-specific heterogeneity, motivating the use of panel data methods in the subsequent analysis.

4. Methodology

4.1 Empirical Strategy

The study investigates the relationship between urbanization and CO₂ emissions using panel data on countries over time. The empirical strategy proceeds in stages, starting with pooled regression models and progresses to add more advanced methods to study trends and biases.

The baseline analysis, as requested in the project description, estimates linear and quadratic relationships between the main regressor and the target. We also explore alternative specifications using best subset selection to choose a parsimonious set of optimally performing variables, and constantly test the subsequent models on this best set to assess whether the choice of control variables affects the results. We then proceed to extend the models by using fixed effects to control for unobserved heterogeneity across countries over time. After conducting baseline analysis, our main goal is to distinguish between

cross-country differences and within-country dynamics, and to assess whether any observed relationships are robust across different model specifications.

In later model specifications, we work with models more appropriate for panel data, since countries differ in unobserved ways independent of time, such as geography, institutions, and industrial structure. If these factors are left unaccounted for, regression models may produce biased estimates by attributing these differences to the explanatory variables. Fixed effects models address this issue by controlling for all time-invariant country characteristics, ensuring that coefficients are identified from within-country variation over time rather than cross-country differences.

To evaluate the model performance, the data is split based on time into training, validation, and testing sets. We choose to do it this way to preserve the time-based structure of the data and to avoid using future observations to predict past outcomes. While predictive performance is assessed using these splits, the primary focus of the analysis remains on inference and interpretation rather than forecasting accuracy.

4.2 The Models

1. Linear Pooled OLS

The first suggested baseline model estimates a linear relationship between CO₂ emissions (y) and urbanization, controlling for GDP per capita and energy use:

$$CO_{2,it} = \beta_0 + \beta_1 Urban_{it} + \beta_2 GDP_{it} + \beta_3 Energy_{it} + u_{it}$$

Where i indexes countries and t indexes time. This model does not account for unobserved country-specific factors.

2. Quadratic Pooled OLS

We proceed to test potential nonlinearity by including a quadratic term in urbanization.

$$CO_{2,it} = \beta_0 + \beta_1 Urban_{it} + \beta_2 Urban_{it}^2 + \beta_3 GDP_{it} + \beta_4 Energy_{it} + u_{it}$$

This allows for a curved relationship and enables computation of a potential turning point in the urbanization-emissions relationship.

3. Fixed Effects

(a) Country FE

The first extension beyond the baseline introduces country fixed effects, which control for time-invariant characteristics per country. This ensures that the estimates capture how within-country changes in urbanization over time affect emissions.

(b) Country + Time FE (The Main Model)

The preferred model uses fixed effects as well, but accounts for both country and time. This model controls for country-specific characteristics and global time trends affecting all countries. As a result, it provides the most reliable estimates, isolating the independent effect of urbanization after accounting for broader structural and global influences.

4.3 Diagnostics and Inference

Residual diagnostics indicate the presence of heteroskedasticity, which means that the variance of errors is not constant across observations. In panel data settings, errors may also be correlated within countries over time, violating standard OLS assumptions. To address these issues, all panel regressions are estimated using clustered standard errors at the country level. This approach provides robust inference by allowing for both heteroskedasticity and within-country correlation of errors.

Multicollinearity is assessed using the Variance Inflation Factor (VIF), which measures how much the variance of estimated coefficients is inflated due to correlation among explanatory variables. The VIF values for the baseline regressors are relatively low, indicating that multicollinearity is not a major concern. This suggests that the chosen variables do not distort coefficient estimates or compromise the reliability of the model.

Failure to account for these issues would result in biased standard errors and unreliable statistical inference.

5. Baseline Model Results — Pooled OLS

5.1 Linear Model

We first estimate a linear pooled OLS model to study the overall, cross-country relationship between urbanization and CO₂ emissions over time. The results of this model indicate that urbanization is statistically significant and is positively associated with CO₂ emissions per capita ($\beta = 0.0524$, p-value = 0.0001). Energy consumption per capita is also positively associated with CO₂ emissions and is statistically significant ($\beta = 0.0001$, p-value < 0.0001), confirming its strong role in driving emissions. The coefficients may seem numerically small, but reflect meaningful proportionality constants when the large units of the quantities involved are considered. GDP per capita is only marginally significant (p-value = 0.0554), which suggests a weaker, non-direct relationship to CO₂ alongside the other two predictors.

These results generally suggest that when all observations are pooled, higher levels of urbanization are significantly associated with higher CO₂ emissions. However, due to observation pooling, this interpretation includes both cross-country and within-country changes, and hence should be interpreted with great caution.

5.2 Quadratic Model

To allow for a potential nonlinear relationship, we modify the linear baseline specification to include a squared term for urbanisation, yielding a quadratic pooled OLS model. This allows for the effect of urbanization on CO₂ emissions to vary across different levels of urbanization, as the squared term grows more rapidly at higher levels.

The results show evidence of nonlinearity, since both the linear and squared urbanization terms are statistically significant (p-values = 0.0009 and 0.0031, respectively). The estimated coefficients of

urbanization and urbanization squared imply the presence of a turning point, estimated at approximately 82.3% urbanization, indicating an inverted-U relationship between urbanization and CO₂ emissions. The turning point lies within the observed range of the data, and hence it may be meaningfully interpreted as the threshold level of urbanization at which CO₂ emissions stop increasing with urbanization and instead begin to be associated with a decrease. Despite the turning point being relevant within our sample, the improvement in model fit compared to the linear specification is modest, implying that the increased complexity only provides limited additional performance and explanation of emission variation.

Overall, while the quadratic model suggests the presence of a nonlinear relationship between urbanization and CO₂ emissions, the same caution as with the linear baseline model must be held. That is, the results should be interpreted with caution as the model is based on pooled observations and may be reflecting cross-country dynamics rather than within-country ones.

5.3 Variable Selection (Best Subset Selection)

After the baseline pooled OLS models, best subset selection was used to find an optimal combination of predictors. Comparisons were made using AIC, BIC, R², RMSE, MAE, and p-values. This yielded many top competing sets of variables with very similar values, and showed that models including energy consumption and population achieve the lowest RMSE and AIC/BIC values. However, the improvement relative to the baseline specification (urbanization, GDP per capita, and energy consumption) is modest (e.g., RMSE $\approx$ 3.54 versus $\sim$ 3.30–3.41 for the top-performing models).

Given the strong theoretical relevance of GDP per capita in explaining CO₂ emissions, particularly in the context of developing countries, we retain the baseline specification for the main analysis to preserve interpretability and parsimony.

As a robustness check, the preferred alternative specification identified through subset selection (urbanization, energy consumption, and population) is also estimated within the fixed effects framework. The results remain consistent, indicating that the main findings are not sensitive to the choice of control variables.

5.4 Diagnostic Checks

Next, we carry out basic diagnostic checks to assess the validity of the pooled OLS models. First, multicollinearity is evaluated using variance inflation factors (VIF), which remain within acceptable ranges: all variables are around or less than 2 in both models, except urbanisation and its square in the quadratic model, which are by construction, going to be trivially highly correlated. Overall, the VIFs indicate no serious multicollinearity among the predictors.

Additionally, a visual inspection of residual plots suggests the presence of heteroskedasticity, as the variance of the residuals appears to change with the value of the predictors. This is consistent with heteroskedasticity commonly observed in panel data, and violates the constant variance assumption of OLS, potentially affecting the reliability of standard errors and p-values.

These findings motivate the subsequent use of clustered standard errors in model fitting and the use of more appropriate panel data methods.

5.5 Summary

Overall, the pooled OLS models suggest the presence of a statistically significant relationship between urbanization and CO₂ emissions, with evidence of nonlinearity and an inverted U-shaped turning point in the quadratic specification. However, the improvement in model fit from the quadratic model is modest, indicating limited gains from introducing nonlinearity. It is important to note that these results are based on pooled observations and hence reflect a combination of both cross-country and within-country changes over time.

Consequently, the observed relationships may be driven by unobserved structural differences across countries rather than purely by the variables in question. This limitation motivates the use of fixed effects models in the next section, which allows us to isolate within-country changes and obtain a more accurate, untainted assessment of the relationship.

6. Fixed Effects Model Results

6.1 Country Fixed Effects

To distinguish between cross-country and within-country relationships, we estimate a country fixed effects model. This model specification controls for the aforementioned unobserved, time-invariant cross-country differences, allowing the coefficients to be identified from strictly within-country variation over time.

The results differ notably from the pooled OLS models. In the fixed effects model, the coefficient of urbanization remains statistically significant, but becomes negative. This suggests that, within a given country, increases in urbanization are associated with decreases in CO₂ emissions per capita when all other variables are held constant. Energy consumption remains positive and highly statistically significant. In contrast to the pooled OLS model, GDP per capita is no longer even marginally statistically significant, which implies that its previously observed effect may have been due to cross-country differences.

Extending the country fixed effects model to include a squared urbanization term yields no statistically significant evidence of nonlinearity, and the implied turning point lies outside the observed range of data, rendering it not meaningful within the sample.

Overall, the newfound change in both the sign and interpretation of the urbanization coefficient underlines the importance of accounting for unobserved country-specific factors when studying the relationship between urbanization and emissions. This motivates the extension of this model further to include time fixed effects in the following section.

6.2 Country and Time Fixed Effects

We now extend the previous specification to include both country and time fixed effects. In addition to controlling for time-invariant differences across countries, this model accounts for global trends that affect all countries over time, such as global shocks or technological advancements.

The results of this model add some additional insight when compared to the previous model. In this specification, the coefficient of urbanization is no longer statistically significant. This suggests that when both country-specific and time-specific effects are accounted for, there is no evidence of a relationship between urbanization and CO₂ emissions within countries over time. Energy consumption per capita remains positive and highly statistically significant, reaffirming its robustness and strength in driving emissions. GDP per capita also continues to be statistically insignificant, which suggests that changes in income over time do not have a significant effect on emissions once other factors are controlled.

Overall, these results suggest that the relationship observed in the pooled OLS models is greatly driven by cross-country changes rather than within-country dynamics. After we account for both country-specific and time-specific effects, urbanization does not appear to have a statistically significant effect on CO₂ emissions. In this case, we fail to reject the null hypothesis that the coefficient of urbanization is equal to zero. This highlights the importance of accounting for both unobserved heterogeneity as well as common time trends when interpreting panel data relationships. These findings also motivate further robustness checks to assess the stability of the obtained results under alternative model specifications.

7. Robustness Checks

To ensure that the main findings are not driven by specific model choices or assumptions, several robustness checks are conducted.

7.1 Variable Selection

Purpose: To assess whether the results depend on the choice of control variables.

Result: As detailed in Section 5.3, alternative specifications, including additional variables such as population, produced only marginal improvements to the model fit, as measured by RMSE and information criteria (AIC/BIC), which were not substantial enough to justify changing the baseline specification. The key result, which was the lack of a statistically significant within-country effect of urbanization, remained unchanged across both the baseline variable and alternate, optimal variable specifications.

7.2 First Differences

Purpose: To examine whether short-run changes in urbanization affect CO₂ emissions by focusing on year-to-year variations within countries.

Result: Urbanization is not statistically significant in the first-difference model, suggesting that short-run changes in urbanization are not associated with changes in CO₂ emissions. This is consistent with the earlier findings from the fixed effects models.

7.3 Log Model

Purpose: To test robustness to functional form and account for skewness in data.

Result: The log-transformed specification's results remain consistent with the fixed effects models. That is, urbanization remains statistically insignificant while energy consumption continues to exhibit a strong positive effect on emissions. This suggests that the earlier main results are not caused by scale effects or skewness in the distribution.

Overall, the findings remain consistent across specifications, providing no robust evidence of a statistically significant within-country relationship between urbanization and CO₂ emissions.

8. Discussion, Heterogeneity, Policy, and Conclusion

The findings of this paper highlight the importance of distinguishing between cross-country and within-country effects. While the pooled models suggest that more urbanized countries tend to have higher CO₂ emissions, the evidence of such a relationship disappears once time fixed effects are accounted for. This suggests that urbanization by itself is not the direct cause of higher CO₂ emissions, and that the relationship observed in the pooled models was only driven by cross-country differences rather than within-country changes over time.

The heterogeneity analysis by GDP and population shows that this result does not differ across the studied country characteristics. This is consistent with the idea that urbanization does not independently drive emissions, but instead may influence other driving factors, such as energy consumption or country-specific factors that are not included in the dataset, but more directly affect emissions. This also helps to explain why the direction and strength of the relationship vary significantly across the countries in the exploratory data analysis.

Across all model specifications, energy consumption per capita remains a strong and statistically significant predictor of CO₂ emissions. This implies that the emissions are more directly and primarily driven by how energy is produced and consumed, rather than by urbanization itself. When this is seen from an economic perspective, this suggests that urbanization may indirectly influence the CO₂ emissions through other factors, such as an increased energy demand or infrastructure development, which are some plausible mechanisms.

These findings have important policy implications. Efforts to reduce CO₂ emissions should focus more on developing ways to improve energy efficiency. That is, countries should implement policies promoting cleaner energy sources, reducing energy consumption, and finding sustainable ways to improve living conditions, rather than just aiming to decrease urbanization. In fact, urbanizations that are well managed through advanced infrastructure and sustainable planning could contribute to the increase of more efficient energy usage and lower emissions.

9. AI Interaction Log Summary — ChatGPT

- “If multiple variable combinations perform similarly, can I keep a more theoretically meaningful baseline model and test alternatives alongside it?”
- “Should population be included as a control variable in addition to urbanization when modelling emissions?”
- “Does fixed effects change slopes across countries or only intercepts?”
- “Why might a theoretically appropriate fixed effects model perform worse than simpler models?”
- “What causes ‘not in index’ errors when working with panel data structures?”
- “Why would a model with time effects perform worse when predicting future periods?”
- “How can I test whether the urbanization effect differs across countries or income groups? Would grouping the data being fitted on work?”

10. Appendix

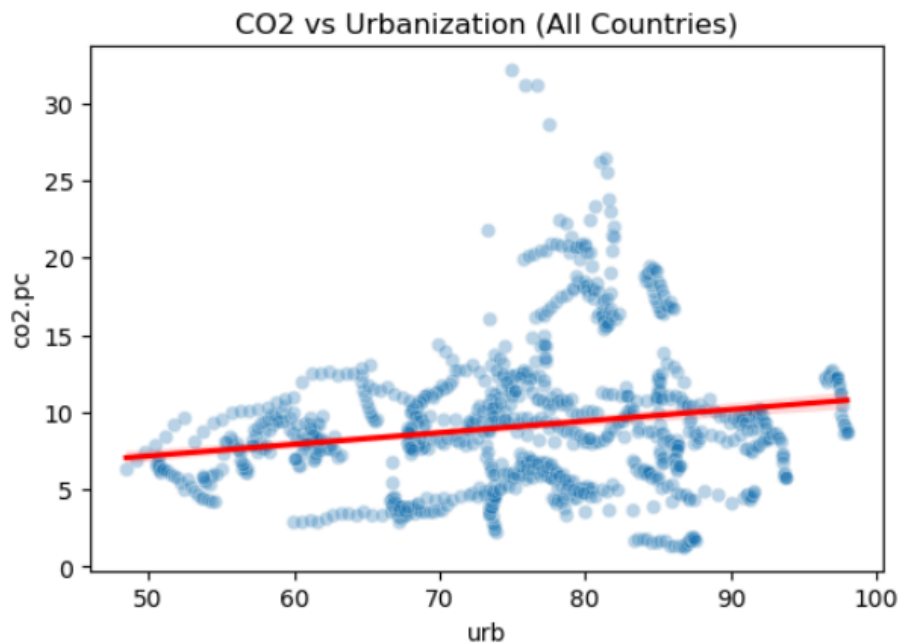


Figure 1

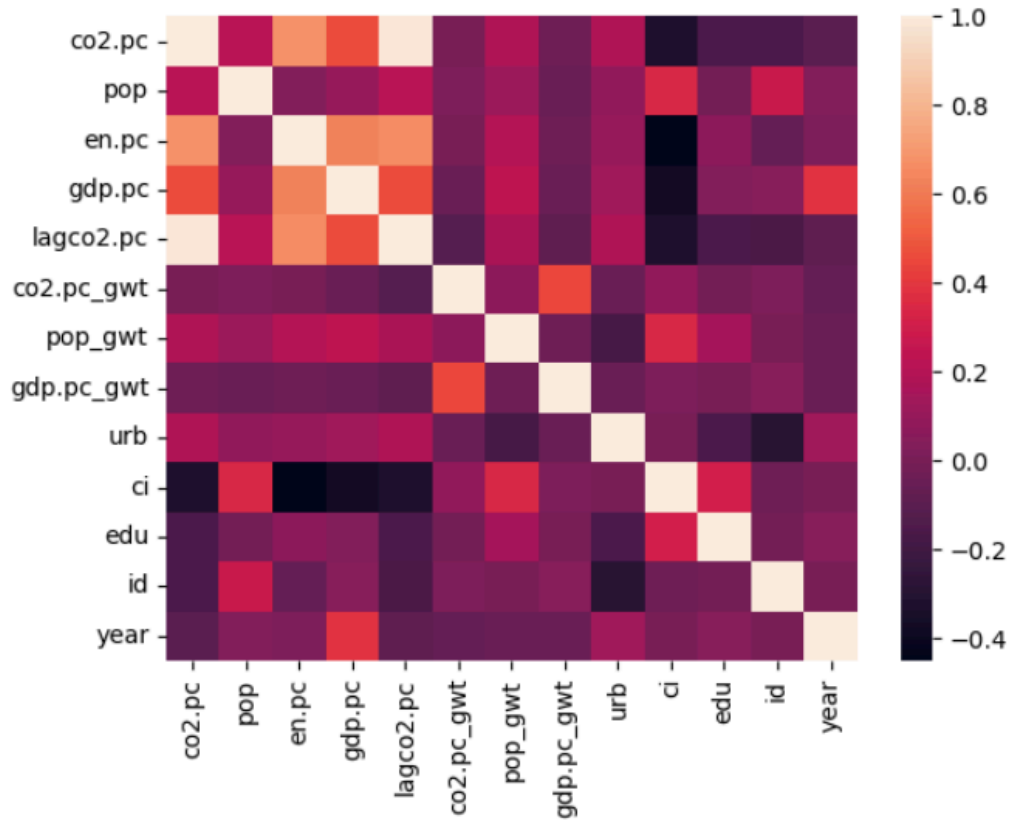


Figure 2

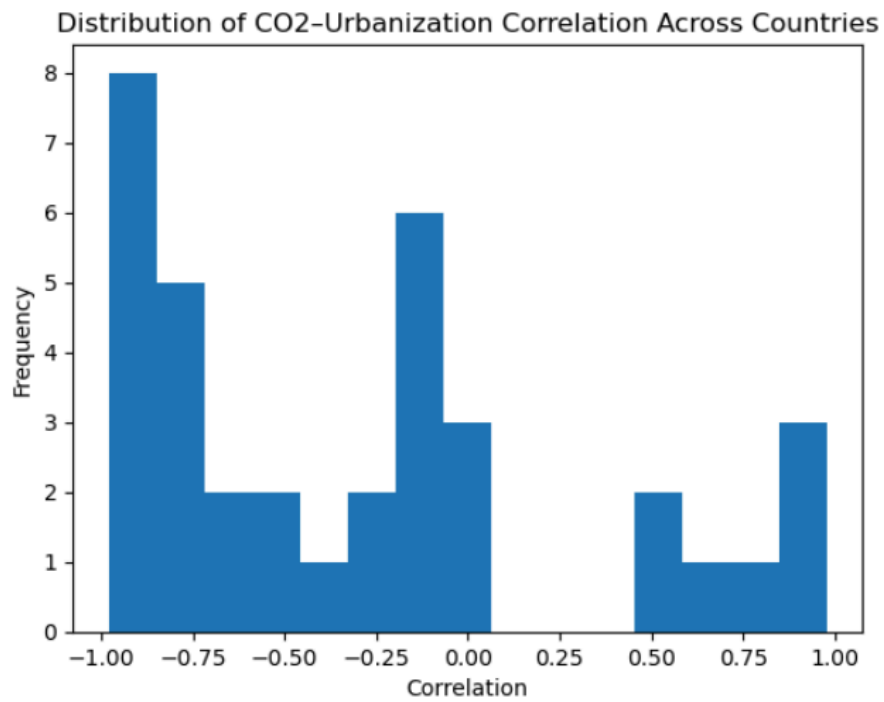


Figure 3

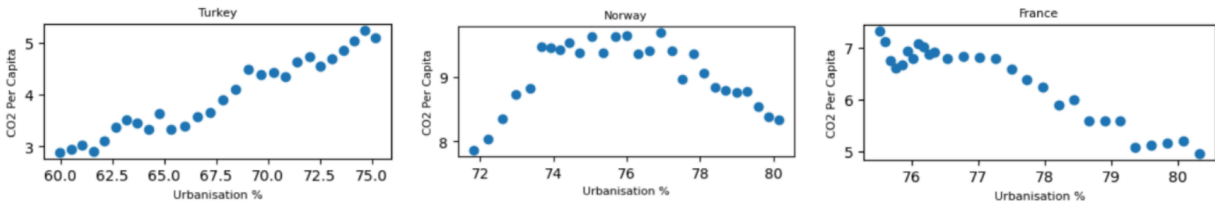


Figure 4 (refer to notebook for all countries)



Figure 5 (refer to notebook for all countries)

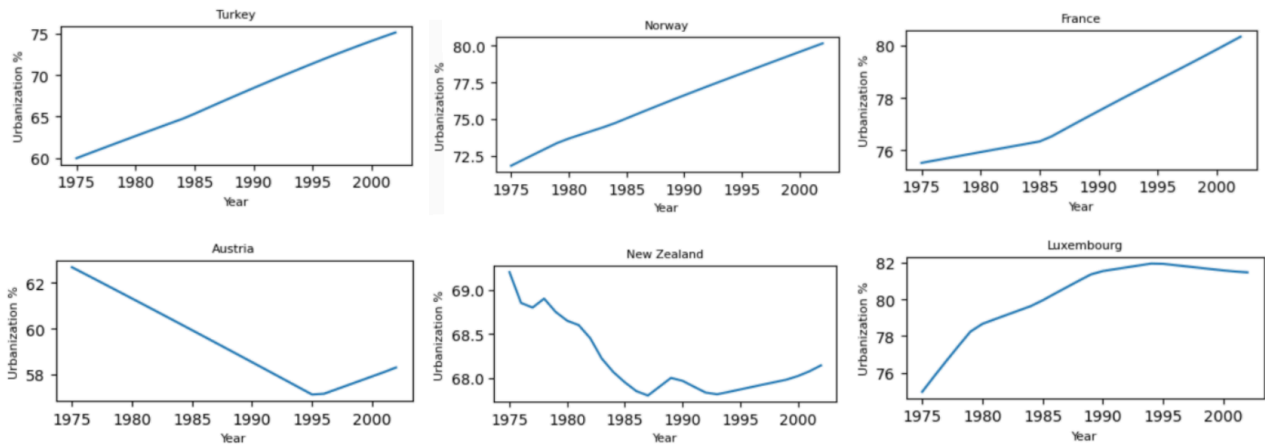


Figure 6 (refer to notebook for all countries)